

scripts/verify-all-constitution-rules.sh — User Guide

Last verified: 2026-05-15 (constitution-compliance plan Phase 1)
Inheritance: HelixConstitution §11.4.32 (Post-Constitution-Pull Validation Mandate) + §11.4.18 (script documentation)

Overview

Master enforcement-engine sweep that wraps every individual constitution-gate + every hermetic test suite into a single invocation. Per §11.4.32 itself: "this gate is the enforcement engine for every other §11.4.x and CONST-NNN rule — without it, new rules cascade as anchors but never get enforced in the codebase".

Mandatory invocation triggers (per §11.4.32):

- Whenever git submodule update --remote constitution produces any content change in the constitution submodule
- Operator-explicit manual invocation
- ci.sh step 5a4 (every CI run)

Usage

```
bash scripts/verify-all-constitution-rules.sh #
    STRICT (default) – exits 1 on any gate failure
bash scripts/verify-all-constitution-rules.sh --strict #
    explicit strict
bash scripts/verify-all-constitution-rules.sh --advisory #
    exits 0 even with failures
bash scripts/verify-all-constitution-rules.sh --json-only #
    suppresses stdout summary; only emits JSON
```

Inputs

Arg	Description
--strict	Default: exit 1 on any gate failure
--advisory	Exit 0 even with failures (for incremental adoption)

Arg	Description
--json-only	Suppress stdout per-gate summary; only write attestation JSON
LAVA_REPO_ROOT env	Override repo root for hermetic testing

Outputs

- **stdout:** per-gate pass/fail summary + final verdict (suppressed by --json-only)
- **.lava-ci-evidence/verify-all/<UTC-timestamp>.json:** structured attestation with sweep metadata + per-gate result list

Attestation JSON schema:

```
{
  "sweep_timestamp": "<ISO timestamp>",
  "sweep_mode": "strict|advisory",
  "sweep_constitution_pin": "<git rev-parse HEAD inside
    constitution/>",
  "total_gates": <int>,
  "pass_count": <int>,
  "fail_count": <int>,
  "all_passed": <bool>,
  "gates": [
    {
      "name": "<gate-name>", "rule_ref": "<$clause-ref>",
      "result": "PASS|FAIL", "duration_seconds": <int>
    }
  ]
}
```

Gates registry (current)

The sweep currently invokes 37 gates across 4 categories (count last revised in Phase 7 — §11.4.25 coverage-ledger added):

1. **Constitution doc parser** (1 gate) — scripts/check-constitution.sh covering §6.A-§6.AE inheritance + §6.W boundary + §11.4.6 no-guessing
2. **Anti-bluff scanners** (9 gates) — non-fatal coverage / **gitignore coverage (Phase 2 — §11.4.30) / nested-own-org-submodules (Phase 5 — §11.4.28, STRICT) / canonical-root-and-upstreams (Phase 8 — §11.4.35 + §11.4.36, STRICT) / helix-deps-manifest (Phase 3 — §11.4.31, STRICT) / coverage-ledger (Phase 7 — §11.4.25, ADVISORY)** / Challenge discrimination Layer 1+2 / Challenge coverage / fixture freshness
3. **No-hardcoding scanners** (3 gates) — UUID / IPv4 / host:port literal scans

4. **Hermetic test suites** (~23 gates) — tests/firebase + tests/ci-sh + tests/compose-layout + tests/tag-helper + tests/vm-* + tests/pre-push/check{4,5,6,7,8,9} + tests/check-constitution/* (now includes test_gitignore_coverage.sh + test_nested_own_org_submodules.sh + test_canonical_root_and_upstreams.sh + test_helix_deps_manifest.sh + test_coverage_ledger.sh)

Per-gate STRICT/ADVISORY note: Phase 5 / Phase 8 / Phase 3 gates have been STRICT-flipped 2026-05-15 after their respective debt closures landed:

- **Phase 5-debt closure:** github cascade merge removed Challenges/.gitmodules (CONST-051(C) flat-layout enforcement); Panoptic no longer nested.
- **Phase 8-debt closure:** 10 install_upstreams scripts landed via Phase 8-debt batch (Challenges, Config, Containers, Discovery, HTTP3, Mdns, Middleware, RateLimiter, Recovery, Tracker-SDK each gained install_upstreams.sh + Upstreams/GitHub.sh + Upstreams/GitLab.sh per §6.W).
- **Phase 3-debt closure:** 16/16 per-submodule helix-deps.yaml manifests landed via Phase 3-debt batch (each authored with HONEST per-submodule deps after analysis).

Phase 7 STRICT-flipped 2026-05-16 after the waiver-backfill agent (commit 20b3fd36) lifted the ledger from 0/20/38 to 48/10/0. The 10 remaining partial rows are honest gaps for structural/glue modules without dedicated unit tests (transitively exercised by downstream features + Challenges). The verifier hard-asserts row-coverage + freshness + schema integrity in STRICT mode; ledger-staleness conditions are the gate's primary failure mode (legitimate gap counts don't trigger the verifier).

The list grows as new constitution gates land. Adding a new gate to the registry requires updating both the sweep + the hermetic meta-test (tests/check-constitution/test_verify_all_rules.sh).

Falsifiability rehearsal (mandated by §11.4.32 itself)

§11.4.32: "sweep's own meta-test (paired mutation §1.1) plants a known violation of each enforced gate and asserts sweep reports FAIL for the planted gate."

Implemented in tests/check-constitution/test_verify_all_rules.sh with 4 fixtures:

- test_clean_tree_passes — sweep passes against the real Lava tree

- `test_gate_failure_propagates` — synthetic minimal repo (no constitution clauses) causes sweep to report `all_passed: false` + `exit 1`
- `test_advisory_mode_returns_zero` — `--advisory` exits 0 even with failures
- `test_attestation_json_structure` — emitted JSON has all required fields

A sweep that exits PASS without running every implementable gate is itself a §11.4.32 violation.

Side-effects

- Creates `.lava-ci-evidence/verify-all/` directory if missing
- Writes one attestation file per invocation (timestamps make file names unique)
- Does NOT modify any tracked files

Cross-references

- HelixConstitution Constitution.md §11.4.32 (the mandate)
- docs/plans/2026-05-15-constitution-compliance.md (Phase 1 — this script)
- docs/helix-constitution-gates.md CM-VERIFY-ALL-CONSTITUTION-RULES row
- tests/check-constitution/test_verify_all_rules.sh (the hermetic meta-test)
- All individual scanners + hermetic suites the sweep wraps